Eshaan Adyanthaya

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Education

- **Vellore Institute of Technology** — Chennai, India
  Integrated M Tech - Software Engineering (3rd Year); GPA: 8.4/10.0 (3.36/4.0)

Experience

- **AI and Robotics Intern - Coratia Technologies** — Remote
  Shark Tank India Featured Startup (66 Crore Funding)
  - **Computer Vision System Development**: Designed and implemented a computer vision-based control system for a 3-degree-of-freedom robotic arm using OpenCV and Python, enabling intuitive gesture-based manipulation.
  - **Real-Time Algorithm Implementation**: Developed real-time hand detection and tracking algorithms to translate hand movements into precise robotic joint commands, achieving seamless human-robot interaction.
  - **Simulation and Hardware Integration**: Created URDF models and RViz simulations for comprehensive testing, then integrated the system with physical hardware using microcontroller-based servo control and inverse kinematics for accurate positioning.
- **Team Manager - Dreadnought Robotics** — Chennai, Tamil Nadu, India
  VIT Chennai
  - **Strategic Leadership**: Lead cross-functional team operations across Programming, Electrical, Mechanical, and Management departments, managing stakeholder relationships and representing the team at competitions.
- **President - Microsoft Innovations Club** — Chennai, Tamil Nadu, India
  VIT Chennai
  - **Executive Leadership**: Lead executive board in organizing technical workshops and hackathons, while developing programs to expand membership and strengthen partnerships with Microsoft and technology organizations.

Projects

- **SGSPose: Neuromorphic-Geometric 6D Pose Estimation**
  Published in IEEE Access (Journal Paper, 2025) — Technologies: SNNs, GNNs, SE(3)-Equivariant Learning, PyTorch
  - **Novel Architecture Development**: Co-authored research paper introducing SGSPose, integrating spiking neural networks, graph neural networks, and SE(3)-equivariant optimization for 6D pose estimation. Achieved state-of-the-art results on 7-Scenes benchmark, reducing mean translation errors by over 70%.
  - **Neuromorphic and Geometric Implementation**: Implemented event-driven SNN encoders with leaky integrate-and-fire neurons for efficient feature extraction, and designed SE(3)-equivariant graph networks preserving geometric consistency under 3D transformations, achieving rotation errors within 8-21° across complex indoor scenes.
- **Project MIRA 2.0 - Autonomous Underwater Vehicle**
  Technologies: Computer Vision, YOLO, MobileNetSSD, ROS, NVIDIA Jetson Xavier, Raspberry Pi
  - **Vision System Development**: Developed image enhancement algorithms using CLAHE for contrast enhancement, Sobel operator for edge detection, and LAB color space processing for improved clarity. Architected object detection pipeline with YOLO and MobileNet-SSD models, creating custom underwater datasets with Roboflow augmentation for real-time marker and obstacle detection in turbid conditions.
  - **System Integration**: Integrated vision systems into AUV control framework for autonomous navigation. Configured NVIDIA Jetson Xavier and Raspberry Pi with Linux and ROS1, managing dependencies for seamless hardware-software integration.
- **4DOF Robotic Arm with Gesture Control**
  Technologies: OpenCV, MediaPipe, Hand Gesture Recognition
  - **Real-Time Gesture Control System**: Developed 4DOF robotic arm controlled through hand gestures using OpenCV and MediaPipe for real-time hand tracking. Implemented dynamic mapping system translating finger joint movements to precise robotic actions with threshold-based actuation for accurate gesture recognition.

Skills Summary

- **Languages**: Python, C, C++, HTML, CSS, SQL, Bash, Java, Arduino, MATLAB
- **Frameworks**: Mediapipe, OpenCV, PyTorch, YOLO, ROS, Roboflow
- **Tools & Platforms**: GIT, Jupyter Notebook, Google Colab, VS Code, PyCharm, MySQL,Linux, Arduino, Raspberry Pi

Publications

- **SGSPose: Neuromorphic-Geometric Fusion for 6D Object Pose Estimation**: IEEE Access (Journal Paper), 2025. Novel architecture integrating spiking neural networks and SE(3)-equivariant learning, achieving state-of-the-art translation accuracy on 7-Scenes benchmark.
- **GOA-UNet: Advanced Metaheuristic Optimization for Medical Image Segmentation**: Measurement Journal, Elsevier, 2025. Automated U-Net hyperparameter tuning using Grasshopper Optimization Algorithm with GAN-based augmentation for crack segmentation, achieving 98.5% accuracy on challenging crack identification tasks.